

Database Management Systems (DBMS)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 21

Question 1 of 21 | GATE 2025 Set 1 (CS1) | Q.15 (Computer Science & IT)

Q.15

A schedule of three database transactions T1, T2, and T3 is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction T_i , $i = 1, 2, 3$. The transaction T1 aborts at the end. Which other transaction(s) will be required to be rolled back?

$R1(X) W1(Y) R2(X) R2(Y) R3(Y) ABORT(T1)$

- (A) Only T2
- (B) Only T3
- (C) Both T2 and T3
- (D) Neither T2 nor T3

Question 2 of 21 | GATE 2025 Set 2 (CS2) | Q.46 (Computer Science & IT)

Q.46

Consider the following relational schema along with all the functional dependencies that hold on them.

$R1(A, B, C, D, E): \{ D \rightarrow E, EA \rightarrow B, EB \rightarrow C \}$

$R2(A, B, C, D): \{ A \rightarrow D, A \rightarrow B, C \rightarrow A \}$

Which of the following statement(s) is/are TRUE?

- (A) R1 is in 3NF
- (B) R2 is in 3NF
- (C) R1 is NOT in 3NF
- (D) R2 is NOT in 3NF

Question 3 of 21 | GATE 2025 Set 2 (CS2) | Q.53 (Computer Science & IT)

Q.53

Consider the database transactions T1 and T2, and data items X and Y. Which of the schedule(s) is/are conflict serializable?

- (A) $R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1)$
- (B) $W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)$
- (C) $R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)$

- (D) W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1) Transaction T1 R1(X) W1(Y) R1(X) W1(X) COMMIT(T1) Transaction T2 W2(X) W2(Y) COMMIT(T2)

Question 4 of 21 | GATE 2025 Set 2 (CS2) | Q.54 (Computer Science & IT)

Q.54

Consider the following relational schema:

Students (rollno: integer, name: string, age: integer, cgpa: real)

Courses (courseno: integer, cname: string, credits: integer)

Enrolled (rollno: integer, courseno: integer, grade: string)

Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?

- (A) SELECT S.name FROM Students S WHERE EXISTS (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno);
- (B) SELECT S.name FROM Students S WHERE SIZEOF (SELECT * FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno) > 0;
- (C) SELECT S.name FROM Students S WHERE 0 < (SELECT COUNT(*) FROM Enrolled E WHERE E.courseno = 1470 AND E.rollno = S.rollno);
- (D) SELECT S.name FROM Students S NATURAL JOIN Enrolled E WHERE E.courseno = 1470;

Question 5 of 21 | GATE 2024 Set 1 (CS1 S5) | Q.21 (Computer Science & IT)

Q.21

In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?

- (A) Only the root node
- (B) All leaf nodes
- (C) All internal nodes
- (D) Only the leftmost leaf node

Question 6 of 21 | GATE 2024 Set 1 (CS1 S5) | Q.44 (Computer Science & IT)

Q.44

The symbol $\rightarrow$ indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

- (A) $(X, Y) \rightarrow (Z, W)$ implies $X \rightarrow (Z, W)$
- (B) $(X, Y) \rightarrow (Z, W)$ implies $(X, Y) \rightarrow Z$
- (C) $((X, Y) \rightarrow Z \text{ and } W \rightarrow Y)$ implies $(X, W) \rightarrow Z$
- (D) $(X \rightarrow Y \text{ and } Y \rightarrow Z)$ implies $X \rightarrow Z$

Question 7 of 21 | GATE 2024 Set 1 (CS1 S5) | Q.46 (Computer Science & IT)

Q.46

Consider the following read-write schedule S over three transactions

T1, T2, and T3, where the subscripts in the schedule indicate transaction IDs:

S: r1(z); w1(z); r2(x); r3(y); w3(y); r2(y); w2(x); w2(y);

Which of the following transaction schedules is/are conflict equivalent to S ?

- (A) T1T2T3
- (B) T1T3T2
- (C) T3T2T1
- (D) T3T1T2

Question 8 of 21 | GATE 2023 Single Session | Q.15 (Computer Science & IT)

Q.15

Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?

- (A) Number of attributes of its relation schema.
- (B) Number of tuples stored in the relation.
- (C) Number of entries in the relation.
- (D) Number of distinct domains of its relation schema. CS

Question 9 of 21 | GATE 2023 Single Session | Q.60 (Computer Science & IT)

Q.60

Consider the following table named Student in a relational database. The primary key of this table is rollNum.

Student

rollNum

name

gender

marks

1

Naman

M

62

2

Aliya

F

70

3

Aliya

F

80

4

James

M

82

5

Swati

F

65

The SQL query below is executed on this database.

```
SELECT *
```

```
FROM Student
```

```
WHERE gender = 'F' AND
```

```
marks > 65;
```

The number of rows returned by the query is

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 10 of 21 | GATE 2022 Single Session | Q.25 (Computer Science & IT)

Q.25

Consider the following three relations in a relational database.

```
(
```

```
,
```

```
),
```

Employee eld Name

```
(
```

```
,
```

```
),
```

Brand bld bName

```
(
```

```
,
```

```
)
```

Own eld bld

Which of the following relational algebra expressions return the set of elds who own all the brands?

(A) $\pi_{eld}(\sigma_{eld = bld}(\pi_{eld, bld}(Own)))$

(B) $\pi_{eld}(\sigma_{eld = bld}(\pi_{eld, bld}(Own)))$

(C) , (() / ()) eld eld bld bld Own Own □ □ □

(D) □ □ (() ()) / () eld eld bld bld Own Own Brand □ □ □ □ □

Question 11 of 21 | GATE 2022 Single Session | Q.31 (Computer Science & IT)

Q.31

Consider a relation

(,
,
,
,
,
)

R A B C D E with the following three functional dependencies.

$AB \rightarrow C$; $BC \rightarrow D$; $C \rightarrow E$;

The number of superkeys in the relation R is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 12 of 21 | GATE 2022 Single Session | Q.56 (Computer Science & IT)

Q.56

Consider the relational database with the following four schemas and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)

Course(cNo, cName, dNo) Register(sNo, cNo)

SQL Query:

SELECT * FROM Student AS S WHERE NOT EXIST

(SELECT cNo FROM Course WHERE dNo = "D01"

EXCEPT

SELECT cNo FROM Register WHERE sNo = S.sNo)

The number of rows returned by the above SQL query is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 13 of 21 | GATE 2019 Single Session | Q.11 (Computer Science & IT)

Q.11 Consider the following two statements about database transaction schedules:

I.

Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.

II.

Timestamp-ordering concurrency control protocol with Thomas' Write Rule can generate view serializable schedules that are not conflict serializable.

Which of the above statements is/are TRUE?

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

Question 14 of 21 | GATE 2019 Single Session | Q.51 (Computer Science & IT)

Q.51 A relational database contains two tables Student and Performance as shown below:

Student

Roll_no.

Student_name

1

Amit

2

Priya

3

Vinit

4

Rohan

5

Smita

Performance

Roll_no. Subject_code Marks

1

A

86

1

B

95

1

C

90

2

A

89

2

C

92

3

C

80

The primary key of the Student table is Roll_no. For the Performance table, the columns

Roll_no. and Subject_code together form the primary key. Consider the SQL query given below:

```
SELECT S.Student_name, sum(P.Marks)
FROM Student S, Performance P
WHERE P.Marks > 84
GROUP BY S.Student_name;
```

The number of rows returned by the above SQL query is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 15 of 21 | GATE 2019 Single Session | Q.55 (Computer Science & IT)

Q.55 Consider the following relations P(X,Y,Z), Q(X,Y,T) and R(Y,V).

P
X
Y
Z
X1
Y1
Z1
X1
Y1
Z2
X2
Y2
Z2
X2
Y4
Z4
Q
X
Y
T
X2
Y1
2
X1
Y2
5
X1
Y1
6
X3
Y3

1
R
Y
V
Y1
V1
Y3
V2
Y2
V3
Y2
V2

How many tuples will be returned by the following relational algebra query?

$\prod(\sigma(P.Y=R.Y \wedge R.V=V2)(P \times R))$

X

$-\prod(\sigma(Q.Y=R.Y \wedge Q.T>2)(Q \times R))$

X

Answer: _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 16 of 21 | GATE 2018 Single Session | Q.12 (Computer Science & IT)

Q.12 Consider the following two tables and four queries in SQL.

Book (isbn, bname), Stock (isbn, copies)

Query 1: SELECT B.isbn, S.copies

FROM Book B INNER JOIN Stock S

ON B.isbn = S.isbn;

Query 2: SELECT B.isbn, S.copies

FROM Book B LEFT OUTER JOIN Stock S

ON B.isbn = S.isbn;

Query 3: SELECT B.isbn, S.copies

FROM Book B RIGHT OUTER JOIN Stock S

ON B.isbn = S.isbn;

Query 4: SELECT B.isbn, S.copies

FROM Book B FULL OUTER JOIN Stock S

ON B.isbn = S.isbn;

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

- (A) Query 1
- (B) Query 2
- (C) Query 3

(D) Query 4

Question 17 of 21 | GATE 2016 Set 1 (CS1) | Q.21 (Computer Science & IT)

Q.21

Which of the following is NOT a superkey in a relational schema with attributes V, W, X, Y, Z and primary key V Y?

- (A) VXYZ
- (B) VWXZ
- (C) VWXY
- (D) VWXYZ

Question 18 of 21 | GATE 2016 Set 1 (CS1) | Q.23 (Computer Science & IT)

Q.23

A database of research articles in a journal uses the following schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE)

The primary key is (VOLUME, NUMBER, STARTPAGE, ENDPAGE) and the following functional dependencies exist in the schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE)

→

TITLE

(VOLUME, NUMBER)

→

YEAR

(VOLUME, NUMBER, STARTPAGE, ENDPAGE)

→

PRICE

The database is redesigned to use the following schemas.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE)

(VOLUME, NUMBER, YEAR)

Which is the weakest normal form that the new database satisfies, but the old one does not?

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

Question 19 of 21 | GATE 2016 Set 2 (CS2) | Q.21 (Computer Science & IT)

Q.21

B+ Trees are considered BALANCED because

- (A) the lengths of the paths from the root to all leaf nodes are all equal.
- (B) the lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
- (C) the number of children of any two non-leaf sibling nodes differ by at most 1.
- (D) the number of records in any two leaf nodes differ by at most 1.

Question 20 of 21 | GATE 2016 Set 2 (CS2) | Q.51 (Computer Science & IT)

Q.51

Consider the following database schedule with two transactions, T1 and T2.

$S = r_2(X); r_1(X); r_2(Y); w_1(X); r_1(Y); w_2(X); a_1; a_2$

where $r_i(Z)$ denotes a read operation by transaction T_i on a variable Z , $w_i(Z)$ denotes a write operation by T_i on a variable Z and a_i denotes an abort by transaction T_i .

Which one of the following statements about the above schedule is TRUE?

- (A) S is non-recoverable
- (B) S is recoverable, but has a cascading abort
- (C) S does not have a cascading abort
- (D) S is strict CS(Set B) 16/18 GATE 2016

Question 21 of 21 | GATE 2016 Set 2 (CS2) | Q.52 (Computer Science & IT)

Q.52

Consider the following database table named water_schemes :

water_schemes

scheme_no

district_name

capacity

1

Ajmer

20

1

Bikaner

10

2

Bikaner

10

3

Bikaner

20

1

Churu

10

2

Churu

20

1

Dungargarh

10

The number of tuples returned by the following SQL query is

```
.  
with total(name, capacity) as  
select district_name, sum(capacity)  
from water_schemes  
group by district_name  
with total_avg(capacity) as  
select avg(capacity)  
from total  
select name  
from total, total_avg  
where total.capacity ≥ total_avg.capacity
```

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]